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The Research on the Construction Method of a Slope Digital Twin Model Driven by Surrogate Models

Fangshi Shi | Wei Yuan | Xinzhan Qin | Wei Wang | Wen Wang | Qinghe Niu | Chao Yin | Meng Wang

School of Civil Engineering, Shijiazhuang Tiedao University, Shijiazhuang 050043, China

Correspondence: Wei Yuan (yuanweisuper001@126.com)

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ABSTRACT

In view of the shortcomings of traditional slope safety monitoring methods, such as low calculation efficiency and poor real-time performance, this paper proposes a digital twin modelling method for slopes based on a surrogate model. Through a stacking integrated learning model combined with a multidimensional error analysis mechanism and adaptive parameter optimisation strategy, a high-precision surrogate model for slope displacement prediction was constructed. Compared with the real displacement data obtained from laboratory experiments and digital image technology (DIC), the prediction error of the optimised surrogate model is reduced from 20% to 5%. In this study, a digital twin system of the Donglingxin landslide was successfully constructed, and the deformation process from 2010 to 2022 was dynamically displayed. The results show that the deformation of the Donglingxin landslide clearly experienced acceleration−stabilisation, and the displacement rate first increased from 12.5 mm/y to 15.5 mm/y and then decreased to 11.8 mm/y, indicating that the landslide entered a relatively stable deformation stage.

1 | Introduction

Landslides are among the most common and destructive geological disasters worldwide. The global spatial distribution of major landslide disasters is shown in Figure 1 [1]. The geological environment in China is complex, especially landslide disasters, which have become among the main types of geological disasters. According to the data of the first national comprehensive risk survey of natural disasters, 132,000 hidden landslide hazards have been identified and registered nationwide, which shows that the distribution of landslide disasters in China is very extensive and that the demand for landslide monitoring and early warning is urgent. Landslides are usually caused by the coupling of a variety of complex factors, such as earthquakes, rainfall and climate change [2], and their occurrence is very concealed and sudden. In addition, their development law is complex, and accurate determination of the early warning threshold is difficult, which presents great challenges for slope stability monitoring and risk prevention and control. Landslides and other geological disasters not only severely damage the surrounding ecological environment but also directly endanger the infrastructure and safety of people's lives and property. Therefore, it is highly important to carry out real-time safety monitoring of slopes and comprehensively and dynamically determine their deformation state and development trend to effectively reduce disaster losses and improve the ability to prevent and mitigate disasters.

Traditional slope safety monitoring technology (such as manual inspection, total station monitoring and displacement metre monitoring) has difficulty addressing the challenges associated with landslide prevention and measurement. In recent years, slope safety monitoring methods have increasingly tended to be digital and intelligent, gradually forming a multidimensional collaborative monitoring method of space (synthetic aperture radar, optical remote sensing, etc.)–space (UAV tilt photography, ground laser scanning, etc.)–Earth (global satellite navigation,

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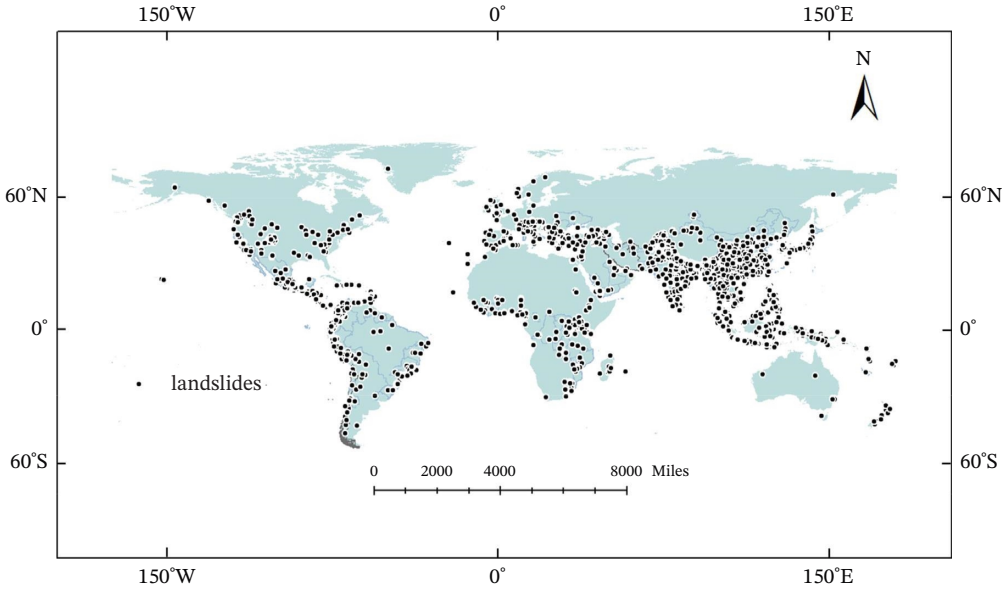


FIGURE 1 | Distribution of the NASA global major landslide database (https://svs.gsfc.nasa.gov/4710).

etc.). Chen et al. [3] proposed a method to evaluate the feasibility of radar interferometry preprocessing and successfully applied it to landslide-prone areas in Guangxi Province, China. Zhang et al. [4] proposed a multimodal SAR optical fusion framework, which successfully captured the early creep displacement of a slope and provided a new idea for landslide monitoring and stability assessment. Wang et al. [5] reconstructed a slope digital elevation model (DEM) through multiple-UAV tilt photography and combined it with an AFSA Elman network to achieve displacement prediction, which proved the efficiency of UAVs in slope monitoring. Wang et al. [6] used unmanned aerial vehicle lidar (UAV lidar) to monitor mine slopes with centimetre accuracy. Wu et al. [7] proposed the MS-CRAIM method by modifying GNSS data, which significantly improved the accuracy of monitoring data. Although emerging monitoring technologies, such as NPR anchor cables [8], microcore pile sensors [9] and magnetic stones, continue to emerge, how to accurately determine the slope state on the basis of monitoring data in a timely manner is still a key problem.

With the rapid development of digital technology, digital twin technology is playing an increasingly important role in slope safety monitoring [10, 11]. The digital twin model of a slope can continuously reflect the actual state of the slope on the basis of real-time monitoring data, which plays an important role in comprehensively understanding the overall situation of the landslide and determining the state of the slope. Dong et al. [12] used displacement monitoring data to directly drive the dynamic quantitative evaluation of slope safety, and the error was controlled within 1%. Ju et al. [13] developed a digital twin model that integrates rainfall data and disaster consequences and performed real-time simulations based on position movement. Shao et al. [14] proposed a method for constructing a dam digital twin platform based on next generation modelling technology (NGM) that could visualise detection results in real time. Wang et al. [15] proposed a method to extract a one-dimensional geometric model from point cloud data and successfully constructed a digital twin model of a long-span structure. To ensure real-time communication between the physical model and the virtual model and to consider the accuracy and efficiency of the model, a surrogate model was introduced into this field, and significant progress has been made [16–18]. Zhou et al. [19] established a digital twin model of the bridge temperature field based on a particle swarm optimisation (PSO) algorithm and realised accurate prediction of the overall temperature field. In addition, the generation countermeasure network (GAN) [20], physical information neural network (PINN) [21] and a variety of stacking algorithms [22, 23] are widely used in the construction of surrogate models, which further promotes development in this direction.

Although digital twin technology based on surrogate models has been widely used in slope safety monitoring, the current research still has the following gaps: (1) in slope safety monitoring, the methods of numerical simulation analysis and monitoring data analysis are relatively mature, but real-time monitoring data and numerical simulations are not integrated for comprehensive monitoring; (2) the prediction accuracy of common surrogate models is often insufficient when dealing with highly nonlinear and complex scenarios of slope deformation and (3) existing models often lack adaptive optimisation mechanisms based on real-time monitoring data. In view of these research gaps, this study proposes a digital twin modelling method for slopes based on a surrogate model and its adaptive optimisation ability, with the aim being to replace expensive numerical simulations with high-precision surrogate models and directly derive the full-field displacement of slopes from limited monitoring data. The main contributions of this paper are as follows:

1. In terms of algorithm development, a novel adaptive surrogate modelling framework is proposed. Unlike traditional static models, this framework integrates a multidimensional error-driven parameter optimisation strategy and a noise-resilient weighting mechanism. These algorithmic improvements enable the model to dynamically self-correct against the "Sim-to-Real" gap, significantly improving prediction accuracy in complex geological environments.

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The average error between the prediction model and the numerical simulation results is 0.02 mm.

2. A digital twin system integrating a physical mechanism and virtual prediction is constructed, and it determines the full-field displacement from the limited monitoring data in real time and resolves the lag of the traditional method.

3. The effectiveness of the method is verified by laboratory tests and engineering cases. After the virtual sensor and actual sensor errors are optimised, the prediction error of the surrogate model at the monitoring point is reduced from 20% to 5%, and the "acceleration–stabilisation" deformation law of the Donglingxin landslide is revealed.

2 | Digital Twin-Driven Framework and Surrogate Model Optimisation Method

2.1 | Digital Twin Collaborative Driving Framework for Slopes

The digital twin model is a system that integrates physics, data and services. The differences between the digital twin model and the numerical simulation model; the pure data-driven model are shown in Table 1. A numerical simulation model is a high-fidelity model based on a physical mechanism that generates high-quality data by solving partial differential equations, but its computational cost is high, and it has difficulty meeting the requirements of real-time analysis. The pure data-driven model directly learns the relationship between the input and output from the monitoring data without the constraint of a physical mechanism. Although its calculation efficiency is high, it strongly depends on the quality and quantity of monitoring data, which contradicts the fact that the number of slope monitoring points is limited. In this study, a numerical simulation model is used to construct and update the high-precision virtual slope, and a pure data-driven model (i.e., a surrogate model) ensures the real-time calculation of the digital twin model.

The framework of the slope digital twin monitoring system proposed in this study is shown in Figure 2. The framework is composed of five core parts: an actual slope, a virtual slope, a slope surrogate model, an Internet of Things (IOT) data acquisition system [24] and a visual display platform. Through the dynamic interaction and efficient integrated processing of multisource heterogeneous data, the system achieves full-range real-time monitoring of the actual state of the slope.

As the physical foundation of the digital twin system, the actual slope contains real geological structure information, such as information related to accurate stratigraphic division, fracture occurrence and rock and soil mechanical parameters. The digitisation of its geometric and physical properties can be achieved by UAV tilt photography, geological section modelling, laser scanning and other technologies. A variety of sensors are arranged inside and on the surface of the slope, including inclinometers, earth pressure boxes, vibration accelerometers, rain gauges and displacement metres, to continuously collect real-time dynamic data such as surface deformation displacement, internal earth pressure, pore water pressure and environmental parameters and provide real-time data input for the construction and updating of virtual slopes.

As the digital mapping of the actual slope, the virtual slope is constructed on the basis of the mechanical parameters of the rock and soil, geological structure characteristics and real-time sensing data. It can dynamically reflect the actual state of a slope and predict and analyse its future behaviour. It is the core of numerical simulation and state deduction.

As a lightweight calculation module in the framework, the slope surrogate model extracts key features from the high-dimensional simulation data generated by the virtual slope based on deep learning technology and constructs a high-precision approximate calculation model. The module has the characteristics of rapid response and low demand for computing resources. It can quickly output stability assessments and early warning information in scenarios where high-fidelity numerical simulations are difficult to run in real time and significantly enhance the real-time performance of the digital twin system.

The IOT system performs data transmission from the actual slope to the digital twin platform and relies on wireless communication technologies such as Lora, 5G, Nb-IOT and edge computing methods to transmit multisource heterogeneous monitoring data to the cloud processing centre in real time. In the process of data integration, the system performs cleaning and fusion to ensure the reliability and consistency of the data and provide stable data support for the dynamic updating of the virtual slope and surrogate model.

As the core interface of human–computer interactions, the visual monitoring platform integrates real-time monitoring data, a three-dimensional virtual model and prediction analysis results. With the help of a three-dimensional rendering engine, the slope state is visually presented in graphical form and supports the visualisation of the risk level, historical data backtracking and

TABLE 1 | Differences among the digital twin model, numerical simulation model and pure data-driven model.

Characteristic dimension	Numerical simulation model	Pure data-driven model	Digital twin model
Essence	Physics-based mechanism model	Statistical-based models	A system that integrates physical entities, data, models, and services
Core foundation	Physical control equations, etc.	Multisource heterogeneous historical monitoring data	Integrates physical mechanisms with real-time monitoring data
Function	High-fidelity simulation, mechanism analysis, and generation of training data	Fast prediction and real-time mapping	Real-time analysis of slope status based on physical mechanisms

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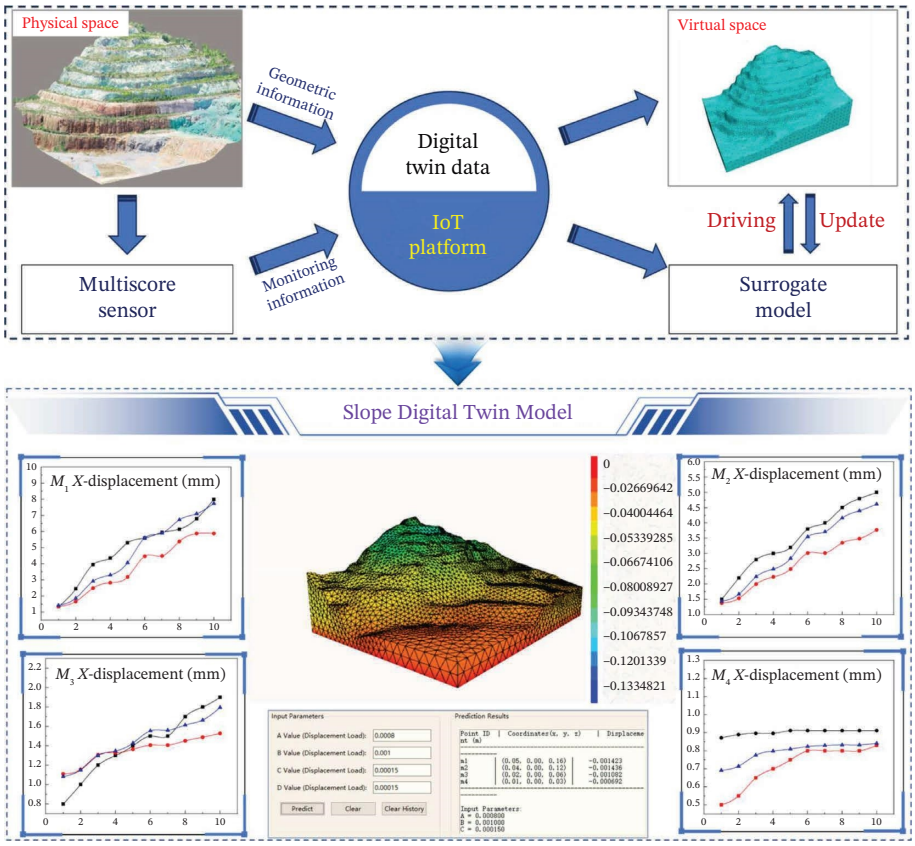


FIGURE 2 | Diagram of the digital twin model.

deformation trend analysis. In addition, the application of augmented reality (AR) and virtual reality (VR) technology has further improved the interactive experience, assisting engineers in intuitively understanding slope safety status and supporting decision-making.

2.2 | Construction of the Heterogeneous Model Fusion Algorithm

In slope safety monitoring, sensors are usually deployed in key areas of potential slip, and their numbers are often limited because of cost constraints. To quickly and accurately retrieve the displacement field of the whole slope on the basis of monitoring information from limited monitoring points, in this study, a stacked integrated algorithm is used to construct a surrogate model for slope displacement calculation, and its process is shown in Figure 3. Stacking is a heterogeneous model integrated learning framework [25]. Its core idea is to integrate multiple types of basic learners and input their prediction results as new features to the meta-learner for secondary training to integrate the advantages of each model, reduce the deviation of basic learners in the prediction, and ultimately output more accurate

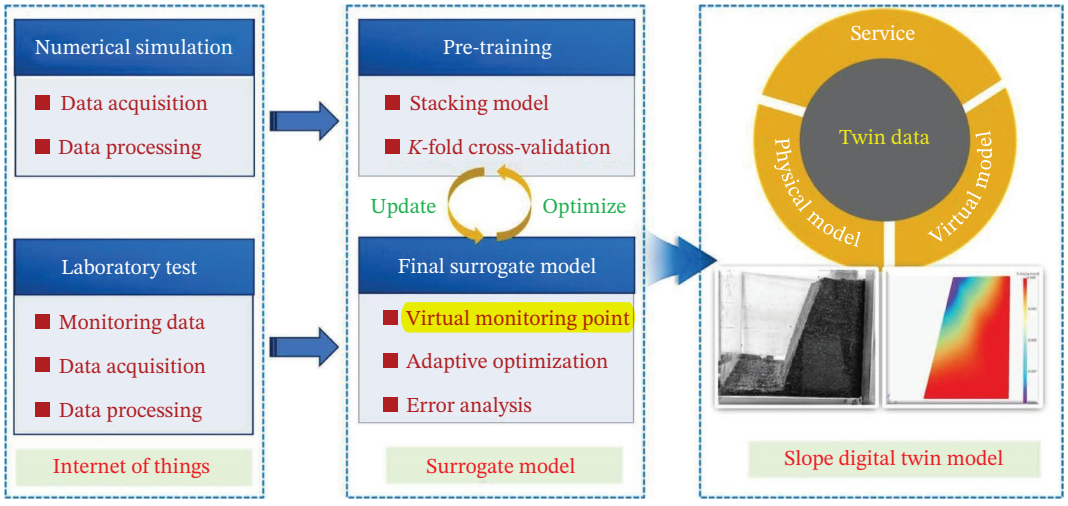


FIGURE 3 | Flowchart of the construction of the slope surrogate model.

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integrated prediction results. Owing to the mechanism of integrating multiple model outputs, the stacking integration algorithm shows good anti-noise and generalisation performance [26, 27].

In stacking ensemble learning, the selection and combination of basic learners have a key impact on the final prediction performance of the model. In accordance with the nonlinear and high-dimensional characteristics of slope deformation and displacement data, five machine learning algorithms, namely, the BP neural network, random forest (RF), extreme gradient lifting (XGBoost), support vector regression (SVR) and radial basis function (RBF) neural network, are preliminarily selected in this study. The BP neural network has strong nonlinear fitting and feature learning ability and is thus suitable for capturing fine patterns in complex data; the RF is insensitive to noise and outliers and has excellent generalisation ability and good robustness; XGBoost can effectively reduce the model deviation through the gradient lifting mechanism, which can significantly improve the prediction accuracy; and on the basis of the principle of structural risk minimisation, SVR can obtain the global optimal and marginal maximum solution; the RBF neural network is good at dealing with local approximation problems, and its physical intuition is strong. It is effectively complementary to the BP neural network in terms of the activation function and structure. This heterogeneous combination of basic learners can ensure that various models produce diverse prediction errors, integrate and correct them through meta-learners, and finally construct a more powerful integrated model. In this study, LightGBM is selected as the meta-learner because it achieves a good balance of prediction accuracy, computational efficiency and generalisation performance and can effectively integrate the output of the basic model to further improve the performance of the overall model.

To evaluate the performance of the model, the commonly used evaluation indicators are the coefficient of determination (R^2), root mean square error (RMSE) and average absolute percentage error (MAPE) [28, 29]. R^2 is used to measure the model's ability to explain the variability of observed data. The closer its value is to 1, the stronger the fitting and regression performance of the model; the RMSE reflects the absolute

TABLE 2 | Calculation expressions for the evaluation metrics.

Evaluation metrics	Calculation formula
Coefficient of determination	$R^2 = 1 - (\sum_{i=1}^{n}(y_i - \hat{y}_i)^2)/(\sum_{i=1}^{n}(y_i - \overline{y_i})^2)$
Root mean square error	$\mathrm{RMSE} = \sqrt{1/n\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$

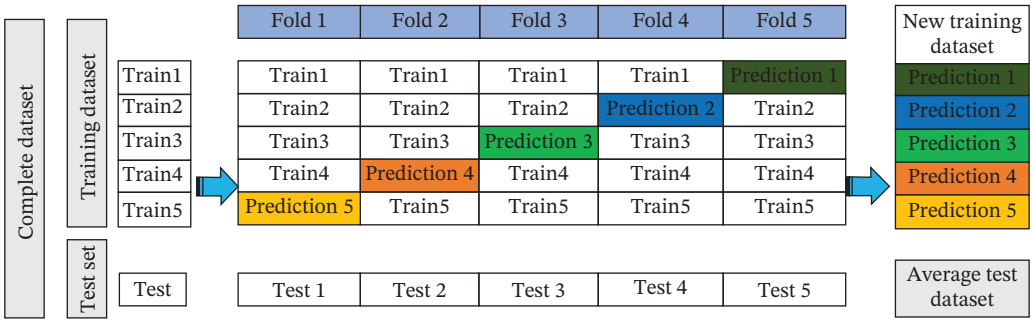


FIGURE 4 | Schematic diagram of 5-fold cross validation.

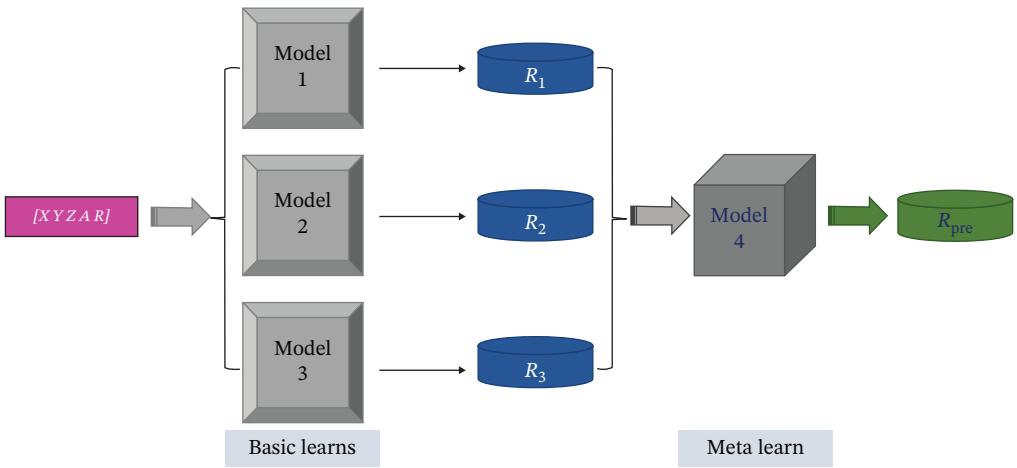


FIGURE 5 | Schematic diagram of the stacking model.

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dispersion between the predicted value and the real value, is more sensitive to large errors, and can effectively reflect the stability of the prediction results. The MAPE evaluates the prediction accuracy from the perspective of relative error and is suitable for comparisons between different levels of data. The lower its value is, the higher the overall prediction accuracy of the model. However, in the prediction of slope deformation, not all measuring points have significant displacement, and the displacement of some nodes is very small or close to zero, which causes the MAPE value to increase sharply or even tend to infinity; thus, the practical significance of the evaluation is lost. Therefore, R^2 and RMSE are ultimately selected as the evaluation criteria for model performance in this study, and their calculation expressions are shown in Table 2.

The stacking heterogeneous integration model is constructed with the following three main steps:

2.2.1 | Data Preparation

Owing to the limited actual slope monitoring data, it is difficult to fully train the surrogate model. In this study, numerical simulation methods were used to generate node displacement data for slopes under various working conditions. The original dataset is represented as [*XYZAR*], where *X*, *Y*, and *Z* are the three-dimensional spatial coordinates of the nodes, *A* is the monitoring information vector, and *R* is the corresponding displacement.

2.2.2 | Basic Learner Training

The 5-fold cross-validation method is used to improve the accuracy of model prediction, and the working principle is shown in Figure 4 [30, 31]. Specifically, the training set is evenly divided into 5 subsets: 1 subset is selected as the validation set, and the remaining 4 subsets are used as the training set; the training is repeated 5 times. Each round of the base learner predicts the validation set and obtains 5 sets of prediction results R_1 to R_5, which are combined into a new prediction feature set $R = \{R_1, R_2, R_3, R_4, R_5\}$, as input to the meta-learner. This process can be executed in parallel to significantly improve computational efficiency.

2.2.3 | Meta-Learner Training

The predicted feature set R_i output by the base learner was used as the input feature of the meta-learner, and it was trained to integrate the advantages of multiple base learners, ultimately forming a stacking ensemble prediction model with higher accuracy and stronger stability. A schematic of the model is shown in Figure 5.

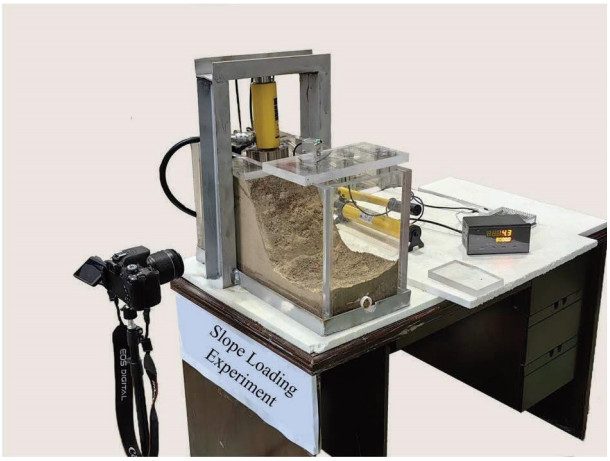


FIGURE 7 | Transparent equipment used to model slope loading.

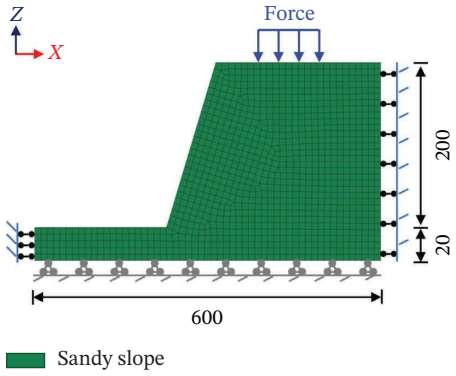


FIGURE 6 | Schematic diagram of the geometric slope model (unit: mm).

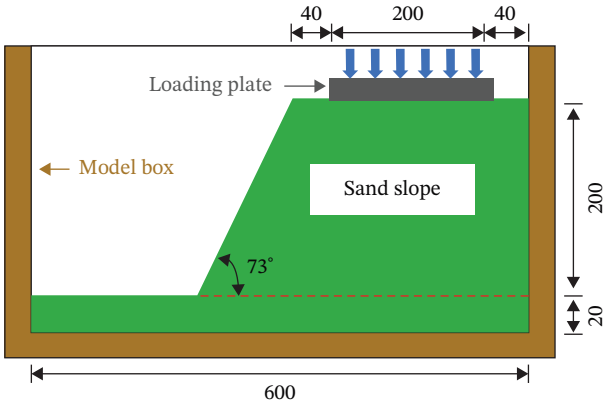


FIGURE 8 | Schematic diagram of model loading (unit: mm).

TABLE 3 | Mechanical parameters of the rock and soil mass of the slope.

Density (kN/m^3)	Elastic modulus (MPa)	Poisson's ratio	Cohesion (kPa)	Friction angle (°)
20	10	0.2	80	30

TABLE 4 | Equipment and sensor models.

Equipment and sensors	Model number
Camera	Canon 5D IV
Camera lens	Canon 24–105 mm lens
Light source	LED fill light
Displacement transducer	Photoelectric sensor BGL-235NZ

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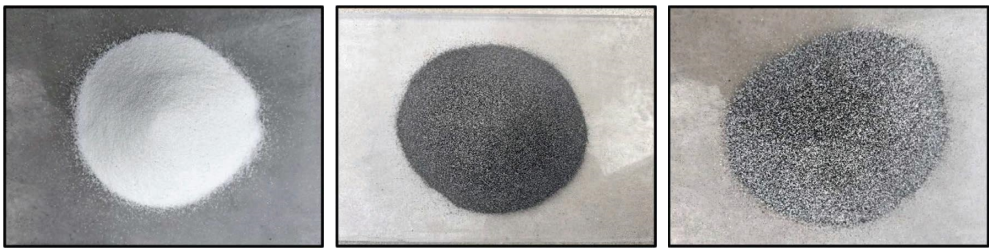
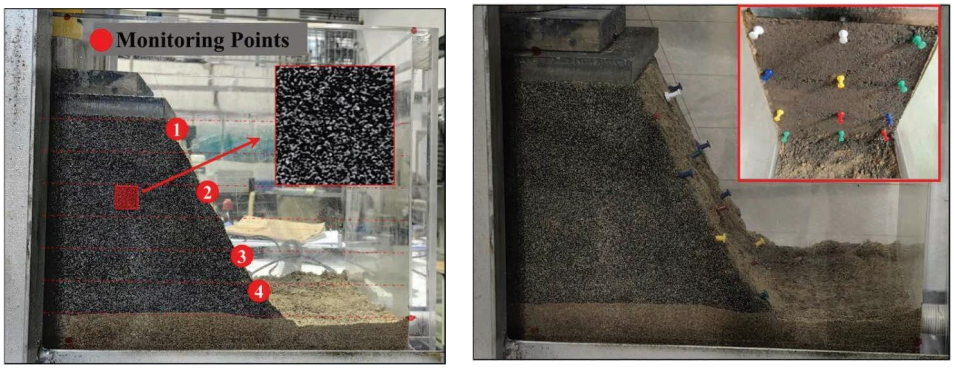


FIGURE 9 | Experimental material. (a) White soil. (b) Black soil. (c) Mixed soil.



(a) (b)

FIGURE 10 | Experimental model. (a) Model side view (b) layout of the monitoring points.

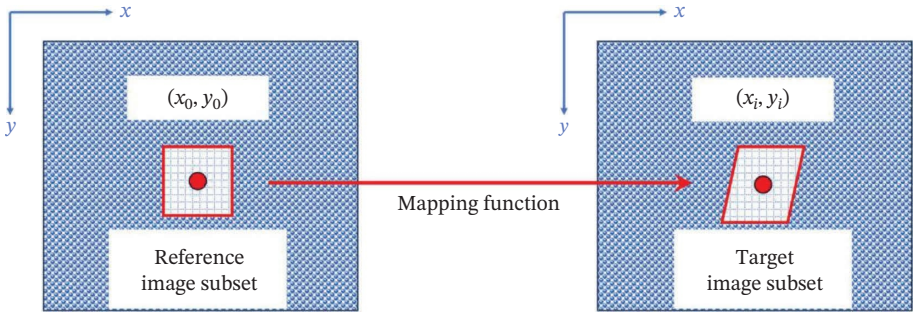


FIGURE 11 | Schematic diagram of the subregion image before and after deformation.

2.3 | Multidimensional Adaptive Optimisation Strategy for Surrogate Models

Owing to the uncertainty of soil parameters and complex environmental conditions, there is often a certain deviation between numerical simulation results and actual slope deformation. To improve the prediction accuracy of the proxy model in practical applications, this study introduces virtual monitoring points into the proxy model and proposes an adaptive optimisation method based on error analysis by comparing the differences between their predicted values and the actual monitoring data at corresponding locations. This method includes three main components: an error feature analysis mechanism, adaptive parameter adjustment, and model updating and retraining.

In error feature analysis, the global error is first statistically analysed from multiple dimensions, including indicators such as the mean error ($error_{mean}$), standard deviation of error ($error_{std}$), and maximum absolute error ($error_{max}$). Second, a relationship

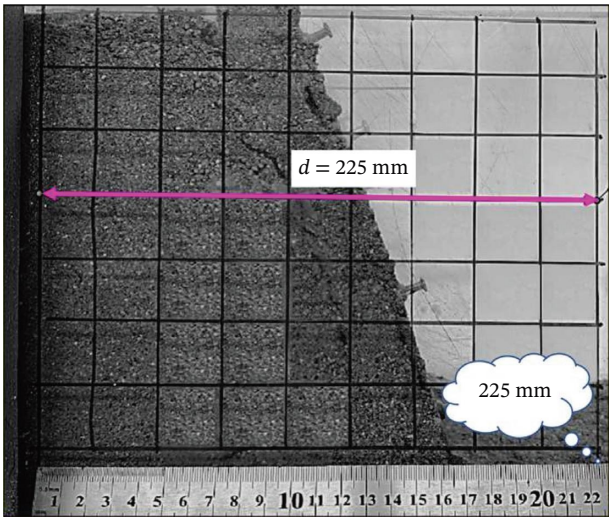


FIGURE 12 | Calibration of the DIC system.

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mapping between the spatial coordinates of monitoring points and prediction errors is established, the spatial patterns of the error distribution are identified, and the characteristics of areas with significant errors are analysed. Finally, a sensitivity analysis of monitoring information is conducted by calculating the correlation coefficients between the data from each monitoring point and prediction errors, evaluating their importance, and identifying key monitoring variables.

In the adaptive parameter adjustment section, a dynamic parameter optimisation strategy is adopted to constrain and adjust key hyperparameters during the model training process to ensure that they are within a reasonable range. The specific adjustment method is shown in formulas (1)–(6):

$$\text{'}n_{\text{estimators}}\text{'} = N \times (1 + \text{error}_{\text{mean}}), \tag{1}$$

$$\text{'max_depth'} = D \times (1 - \text{error}_{\text{std}}), \tag{2}$$

$$\text{'learning_rate'} = \alpha \times (1 - \text{error}_{\text{mean}}), \tag{3}$$

$$\text{'min_samples_split'} = S \times (1 + \text{error}_{\text{std}}), \tag{4}$$

$$\text{'subsample'} = \min(1.0, 0.8 + \text{error}_{\text{std}}), \tag{5}$$

where N is the number of basic estimators, D is the depth of the tree, α is the learning rate and S is the splitting threshold.

Second, a sample weighting strategy based on prediction error was introduced, which assigns lower weights to samples with larger errors to reduce their negative impact on model training and improve the prediction accuracy of the model in key areas. The method used to update weights is shown in the following equation:

$$w = \frac{1}{(1 + |y_p - y_t|)}, \tag{6}$$

where w is the weight corresponding to the sample, y_p is the predicted value of the model and y_t is the test data value.

Finally, the original surrogate model is updated and retrained on the basis of error analysis to achieve dynamic optimisation of model performance. The optimisation process focuses not only on improving the global prediction accuracy but also on improving the prediction accuracy of local areas to ensure that the model has better adaptability and reliability in complex slope environments.

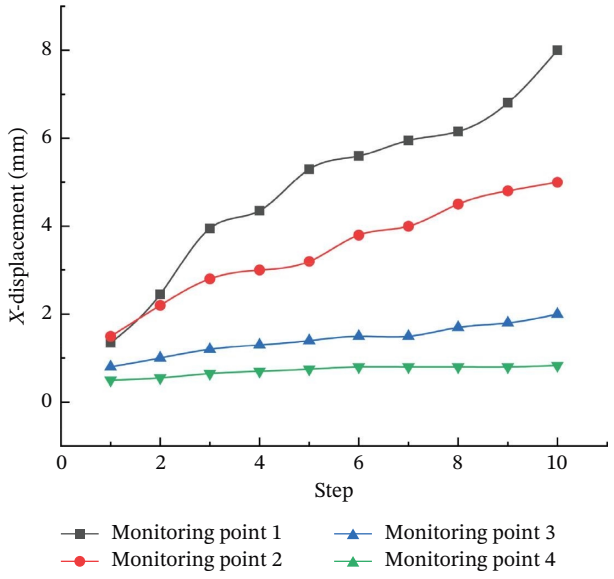


FIGURE 13 | Horizontal displacement of monitoring points.

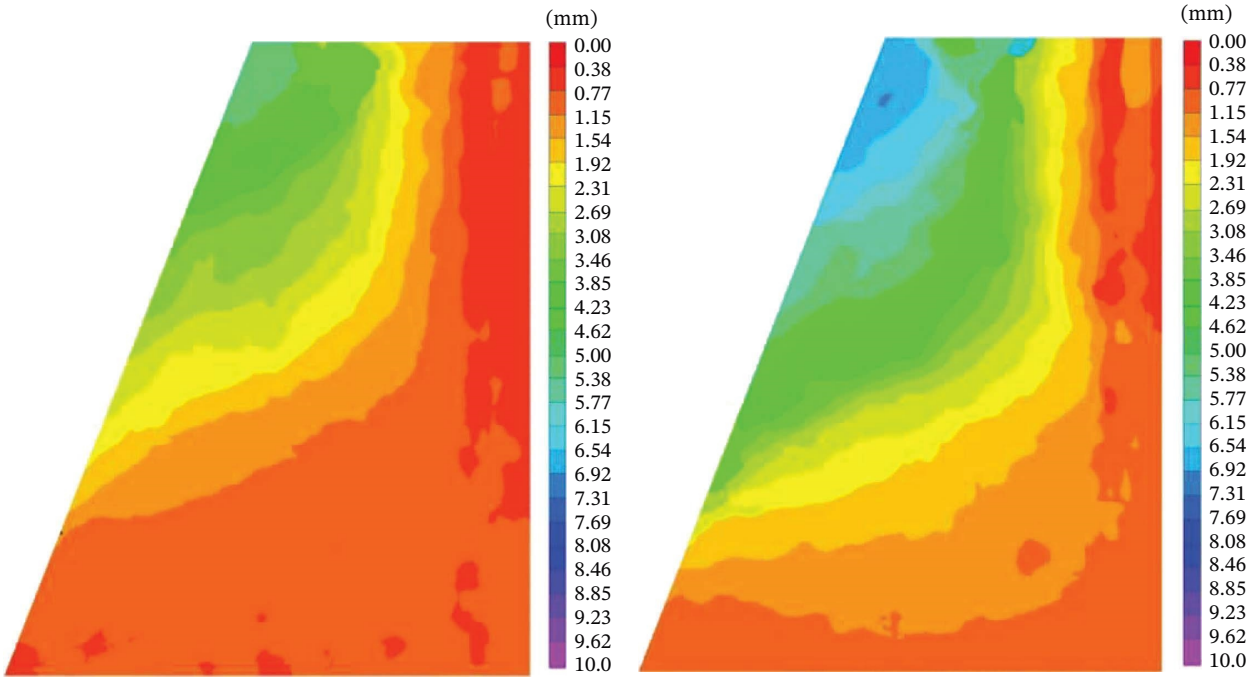


FIGURE 14 | (Continued)

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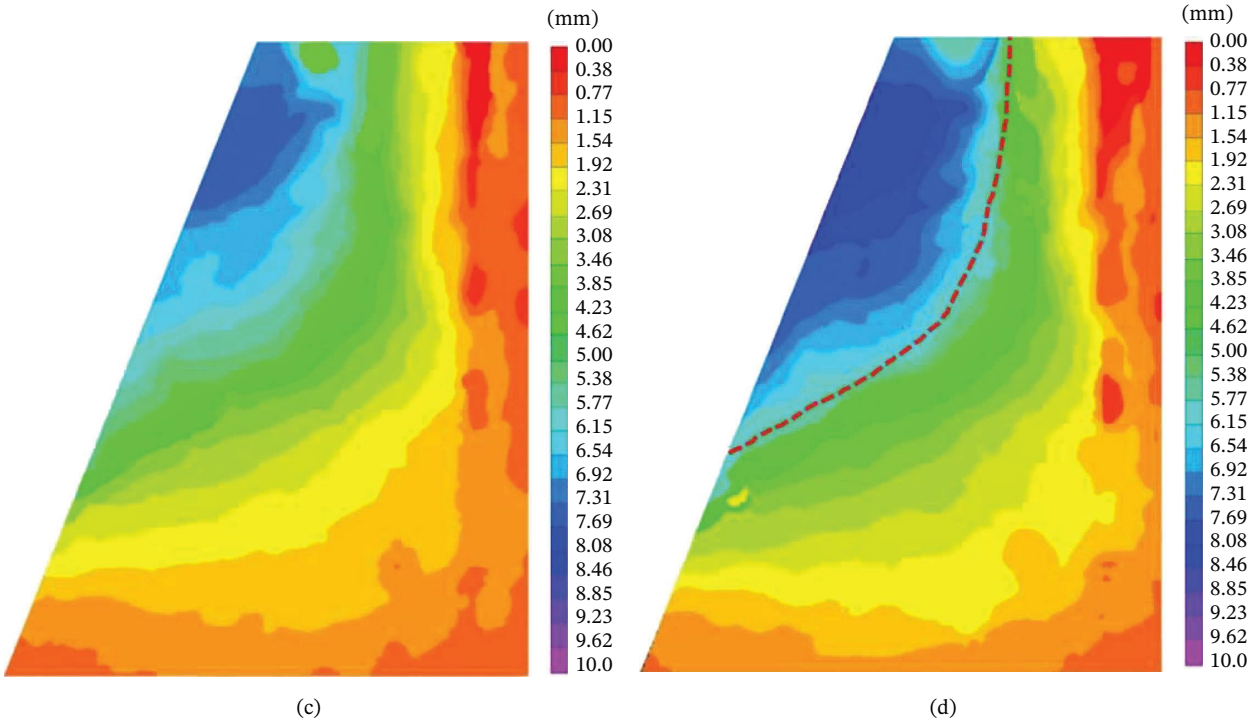


FIGURE 14 | Cloud map of the horizontal displacement and deformation of the slope. (a) Displacement cloud map of step 1. (b) Displacement cloud map of step 2. (c) Displacement cloud map of step 3. (d) Displacement cloud map of step 4.

3 | Verification and Materials

3.1 | Numerical Simulation

To expand the training samples to compensate for the limited amount of actual monitoring data, in this study, a systematic numerical simulation analysis is performed, and the node displacement data for the slope are obtained under all working conditions. The finite difference method is used for the simulation, and the geometric model and grid division of the slope are shown in Figure 6. The Mohr–Coulomb criterion is selected as the constitutive model. The mechanical parameters of the rock and soil mass are listed in Table 3.

3.2 | Laboratory Experiments

A self-developed loading frame is used in the test. The dimensions of the model box are $600 \times 200 \times 300$ mm (length × width × height), and the material is a 20 mm thick transparent acrylic plate, which is convenient for observing and recording the slope deformation process. The real object is shown in Figure 7 [32, 33]. The bottom of the model box is fixed on the loading frame, and the load is applied to the top through the jack. The dimensions of the loading plate are $200 \times 200 \times 20$ mm, and the loading diagram is shown in Figure 8. To reduce the boundary friction effect, the inner wall of the model box is coated with a lubricating oil layer. During the test, the displacement sensor is used to monitor the displacement of the measuring point, and the loading pressure value is recorded in real time by the pressure acquisition instrument. The slope deformation image is continuously collected by a camera. To avoid interference from natural light, fill lights are used to provide stable lighting during shooting. The equipment and sensor models are shown in Table 4.

3.3 | Experimental Material

To prepare the speckle field suitable for digital image correlation (DIC) analysis, the slope soil is dyed black and white and then mixed evenly at a ratio of 1:1, and the moisture content of the mixed soil

TABLE 5 | Model evaluation results.

Base learners	Training set		Validation set		Test set	
	R^2	RMSE	R^2	RMSE	R^2	RMSE
RF	0.935	2.503	0.911	2.501	0.903	2.877
XGBoost	0.941	1.965	0.927	2.076	0.906	2.431
BP	0.799	2.556	0.778	2.951	0.757	3.674
SVR	0.783	3.141	0.754	3.338	0.712	4.009
RBF	0.636	3.855	0.612	3.992	0.599	4.012
Stacking	0.971	1.511	0.955	2.007	0.952	2.108

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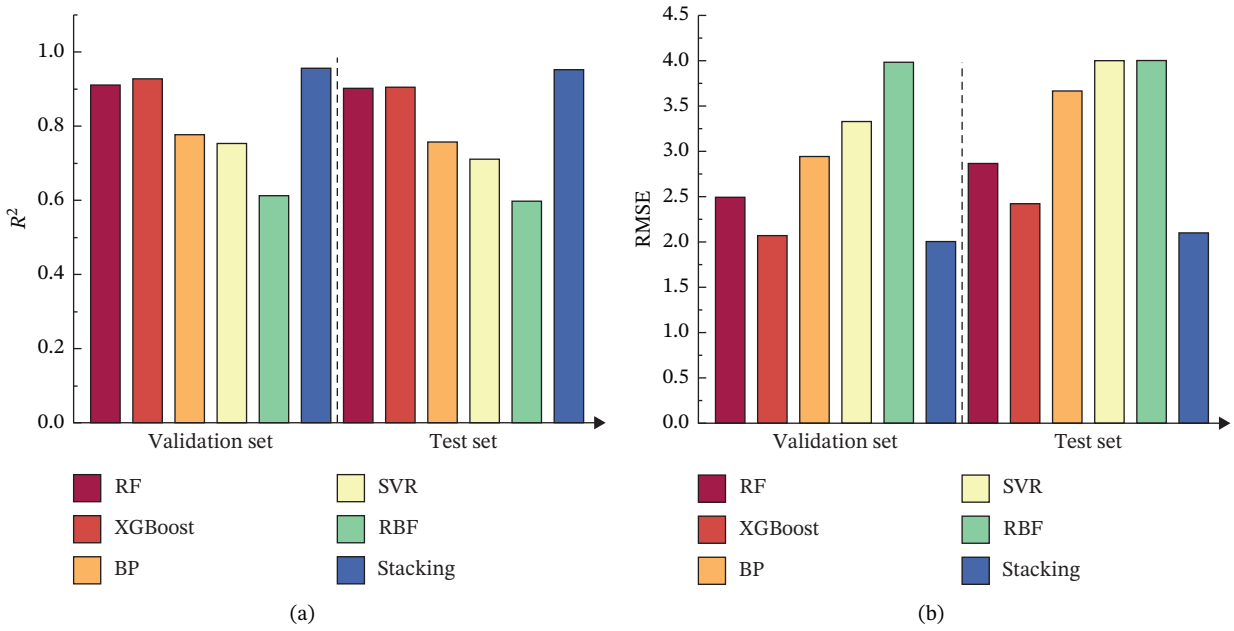


FIGURE 15 | Comparison of the model evaluation results. (a) R^2. (b) RMSE.

was controlled to 3%. The material preparation process is shown in Figure 9. The white soil is shown in Figure 9(a), the black soil is shown in Figure 9(b), and the mixed speckle soil sample is shown in Figure 9(c). When filling, the model box is placed in layers, with 50 mm for each layer, and compacted layer by layer, as shown in Figure 10(a). A total of 12 monitoring points are arranged on the slope surface, as shown in Figure 10(b). A laser displacement sensor is used to monitor the horizontal displacement of each point, and finally, the test slope model is formed. The coordinates of the corresponding monitoring points in the numerical model are set as follows: M_1 (50, 0, 160), M_2 (40, 0, 120), M_3 (20, 0, 60), and M_4 (10, 0, 30). To avoid distortion of the speckle pattern due to excessive deformation, the maximum allowable displacement of the slope is determined to be 10 mm throughout the test, and the deformation range is strictly controlled during the test.

3.4 | DIC System Principle and Calibration

DIC calculates the displacement field by determining the change in the position of the image subarea, which includes mainly the whole-pixel displacement search and subpixel displacement calculation. This study uses the inverse combination Gaussian Newton (IC-GN) algorithm based on the zero normalised least squares difference (ZNSSD) criterion to achieve stable and efficient tracking of subpixel displacement. A subregion diagram before and after deformation is shown in Figure 11. Through the pixels in the subregion, the cross-correlation function is constructed, the nonlinear least squares solution is obtained, and the displacement and other data are calculated. The similarity between the reference subset and the target subset is calculated using the normalised error square sum correlation coefficient, which is expressed as follows:

$$C_{\mathrm{ZNSSD}}(x_0, y_0, x_i, y_i) = \frac{\sum F(x_0, y_0) \cdot G(x_i, y_i)}{\sqrt{\sum F(x_0, y_0)^2 \cdot G(x_i, y_i)^2}}, \tag{7}$$

where $F(x_0, y_0)$ is the grey value of the reference image subset point (x_0, y_0) and $G(x_i, y_i)$ is the grey value of the corresponding deformation subset point (x_i, y_i). $C = 1$ indicates complete correlation, whereas $C = 0$ indicates no correlation.

There is a linear first-order transformation between the pixel displacement of the changed subregion and the pixel displacement of the corresponding subregion in the reference image, and its coordinate calculation expression is as follows:

$$x_i = x_0 + u + \frac{\partial u}{\partial x}(x_i - x_0) + \frac{\partial u}{\partial y}(y_i - y_0), \tag{8}$$

$$y_i = y_0 + v + \frac{\partial v}{\partial x}(x_i - x_0) + \frac{\partial v}{\partial y}(y_i - y_0), \tag{9}$$

where u and v are the x-direction displacement and y-direction displacement of the reference subset, respectively.

To convert the pixel displacement obtained by DIC analysis into the actual displacement and ensure the accuracy of the measurement results, the DIC system is calibrated. A ruler marked with a true scale (225 mm) is placed on the observation surface of the slope model, as shown in Figure 12. The scale factor (pixels/mm) of the system was determined by calculating the pixel span of the calibration target in the image. All pixel displacements were subsequently converted to physical displacements using this factor.

TABLE 6 | Hyperparameter optimisation results.

Model	Hyperparameters	Optimisation results
RF	min_samples_split	3
	min_samples_leaf	1
	n_estimators	500
	max_depth	23
XGBoost	learning_rate	0.193
	max_depth	11
	n_estimators	100
BP	hidden_layer_sizes	One hidden layer (100)
	learning_rate	0.0002

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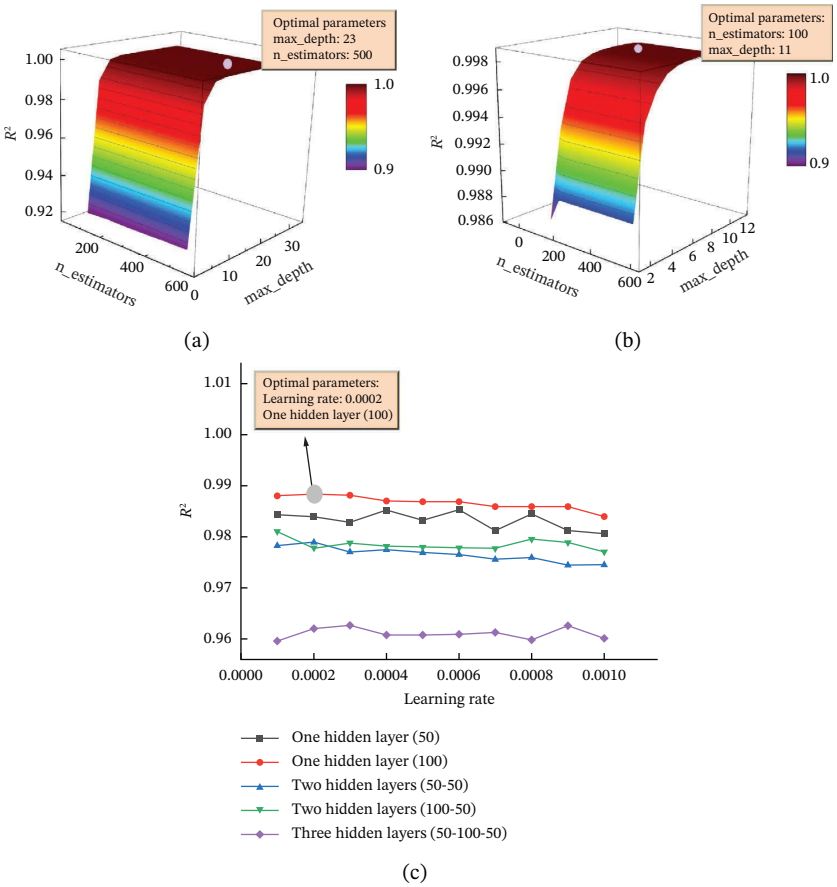


FIGURE 16 | Sensitivity analysis diagram of hyperparameter optimisation. (a) RF. (b) XGBoost. (c) BP.

4 | Discussion of the Results

4.1 | Slope Loading Failure

Under the load of the top jack, the slope gradually deforms with increasing load, and the horizontal displacement continues to increase. The change in horizontal displacement at each monitoring point is shown in Figure 13. The displacement of the four monitoring points increases with increasing load, and the horizontal displacement of monitoring point 1 at the top of the slope is the greatest, and that of monitoring point 4 at the bottom of the slope is the smallest. When the horizontal displacement of point 1 is close to 8 mm, the displacements of points 2, 3 and 4 are

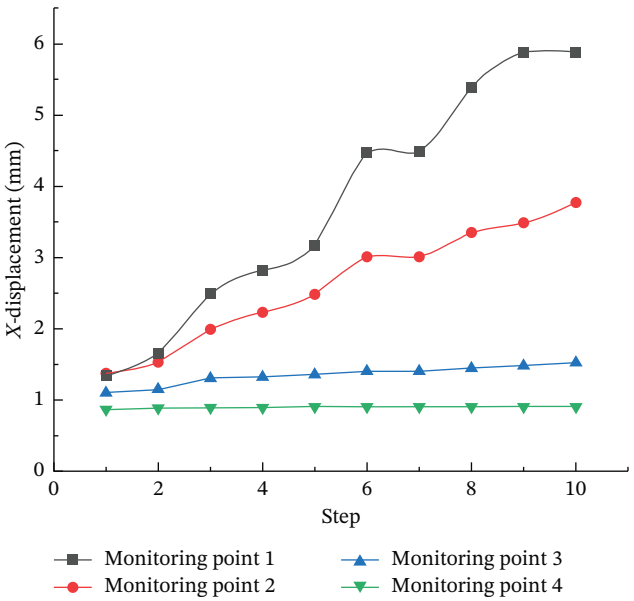


FIGURE 17 | Prediction results of presurrogate model.

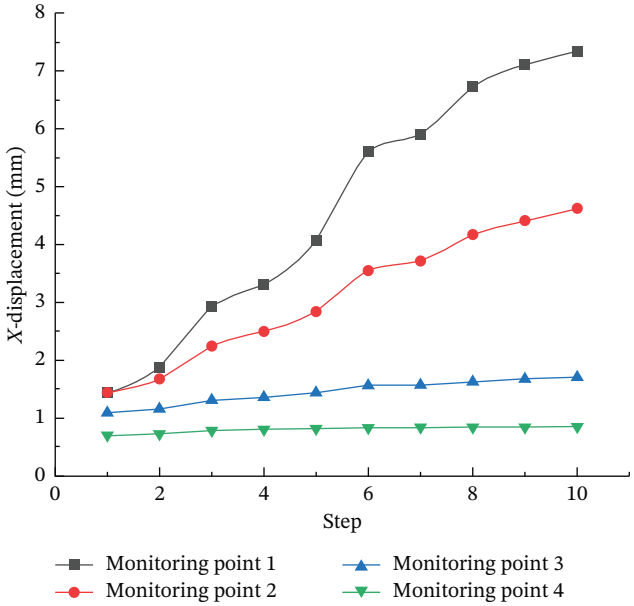


FIGURE 18 | Prediction results of the optimised surrogate model.

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approximately 5, 1.8 mm and less than 1 mm, respectively, indicating that the slope deformation gradually attenuates from the top down, which conforms to the typical law of slope deformation.

Through DIC analysis of slope deformation images, the horizontal displacement nephogram for a slope under different loading stages is obtained, and the results are shown in Figure 14. With increasing vertical load, the horizontal displacement of the slope gradually increases, and an obvious sliding surface can be seen in Figure 14(d). The cloud map shows that the maximum displacement is approximately 8.3 mm, which is very close to the 8.0 mm displacement measured at monitoring point 1, and the absolute error is only 0.3 mm, indicating that the DIC analysis results are highly accurate and are in good agreement with the actual monitoring data.

4.2 | Stacking Model Selection

To quantitatively compare the performance of each basic model and integrated stacking model, the training data are divided into a training set, a validation set and a test set according to

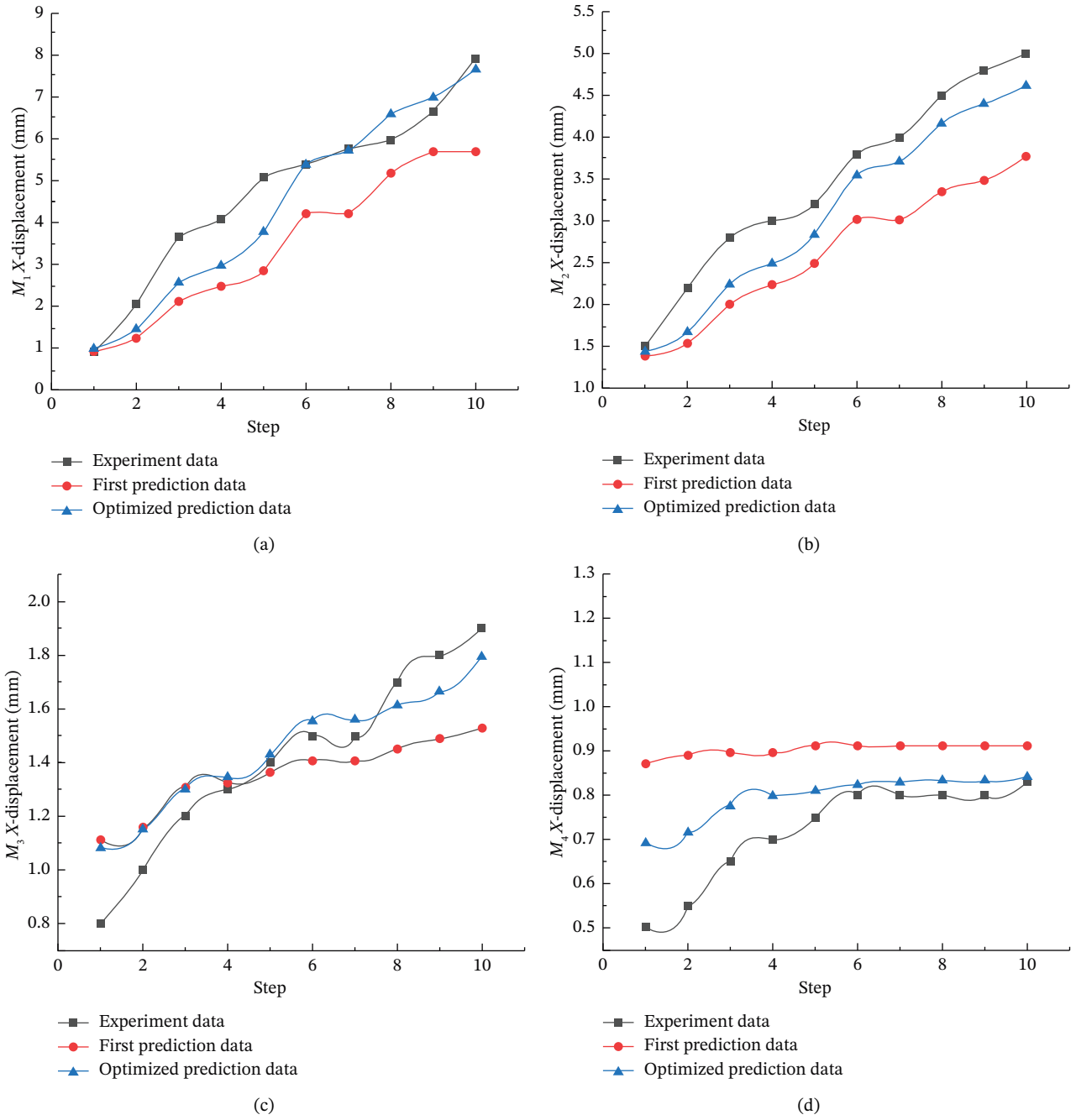


FIGURE 19 | Comparison of the horizontal displacements of the monitoring points. (a) *X*-displacement of monitoring point M_1. (b) *X*-displacement of monitoring point M_2. (c) *X*-displacement of monitoring point M_3. (d) *X*-displacement of monitoring point M_4.

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proportions of 70%, 15% and 15%, respectively. Table 5 lists the specific results of all the models on the evaluation index. In the training set and validation set, XGBoost performs best in terms of R^2 and RMSE among the five basic models; the overall prediction performances of the RF, XGBoost and BP neural network models are the best. On the test set, although the performances of the above three models decrease slightly, they are still significantly better than the SVR and RBF models are, indicating good generalisability. Considering the R^2 and RMSE, the RF, XGBoost and BP neural network models are selected as the base learners of the integrated stacking model. The training results show that the stacking model is superior to all the single basic models in terms of R^2 and RMSE, indicating better prediction accuracy and stability. The evaluation results for the verification set and test set of each base learner and stacking model are shown in the histogram in Figure 15. The R^2 values of the models are compared, as shown in Figure 15(a), and the RMSE values of the models are compared, as shown in Figure 15(b).

4.3 | Hyperparameter Optimisation Strategy and Sensitivity Analysis

To improve the accuracy of the model, random search is selected to optimise the hyperparameters, and the objective of optimisation is to minimise R^2. The optimisation results are shown in Table 6. In the RF model, the minimum number of segmented samples is 3, the minimum number of leaf samples is 1, the number of trees is (100, 600), and the maximum depth of trees is (1, 29). In the XGBoost model, the learning rate is 0.193, the number of trees is (100, 600), and the maximum depth of the trees is (1, 12). In the BP model, five kinds of hidden layer structures are selected: one hidden layer (50), one hidden layer (100), two hidden layers (50-50), two hidden layers (100-50), and three hidden layers (50-100-50).

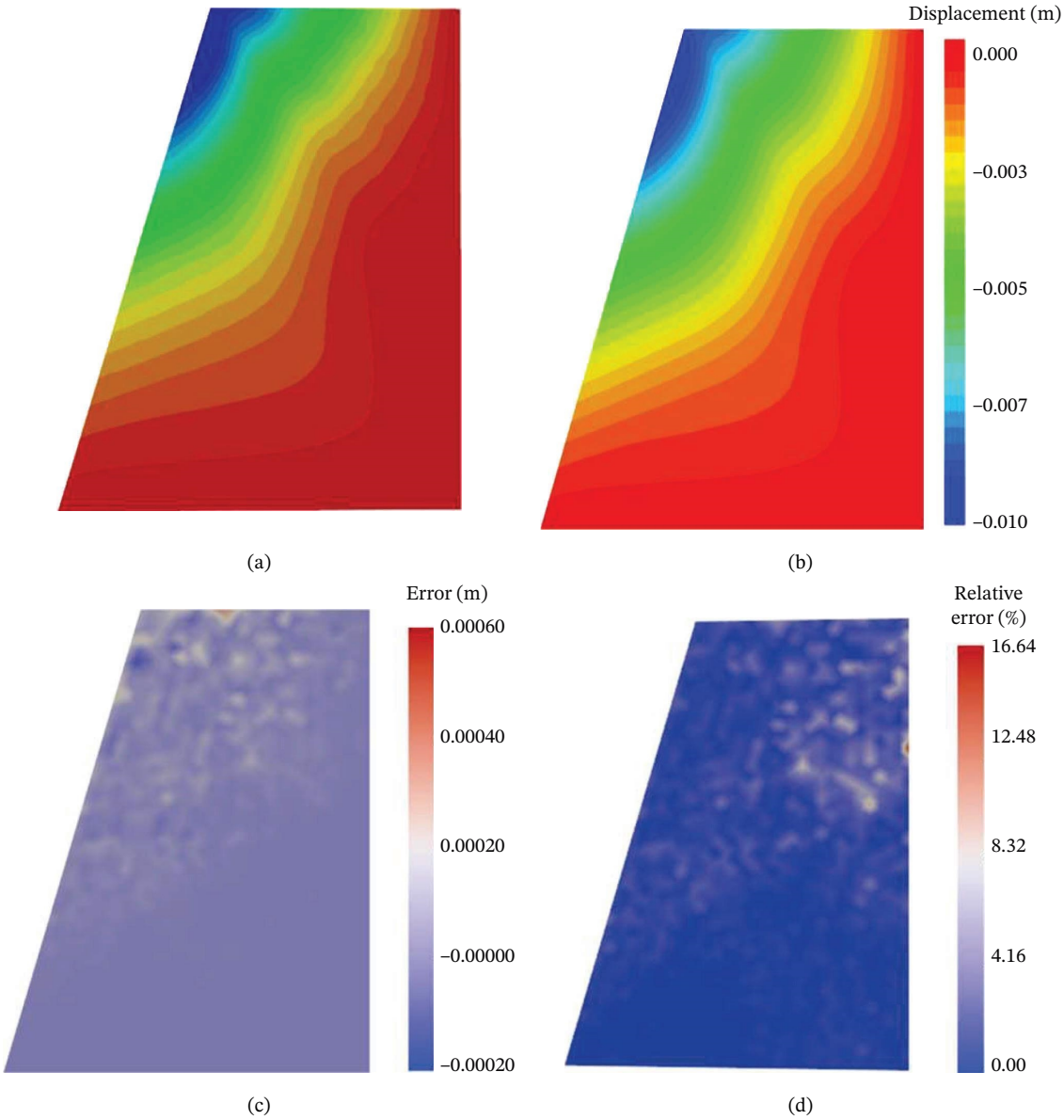


FIGURE 20 | Comparison between the prediction results of the surrogate stacking model and the numerical simulation results. (a) Surrogate stacking model prediction results. (b) Numerical simulation results. (c) Error cloud map for the surrogate stacking model. (d) Relative percentage error cloud map.

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The hyperparameter analysis results of the three models are shown in Figure 16. In the RF and XGBoost models, R^2 increases significantly with increasing max_depth, indicating that max_depth is highly sensitive, and R^2 changes slightly with n_estimators. The max_depth is selected as 23 for the RF model and 11 for the XGBoost model. For the BP model, when the hidden layer structure is one hidden layer (100), R^2 is better than that of the other hidden layer structures; R^2 increases first but then decreases with learning, and the optimal learning rate is 0.0002.

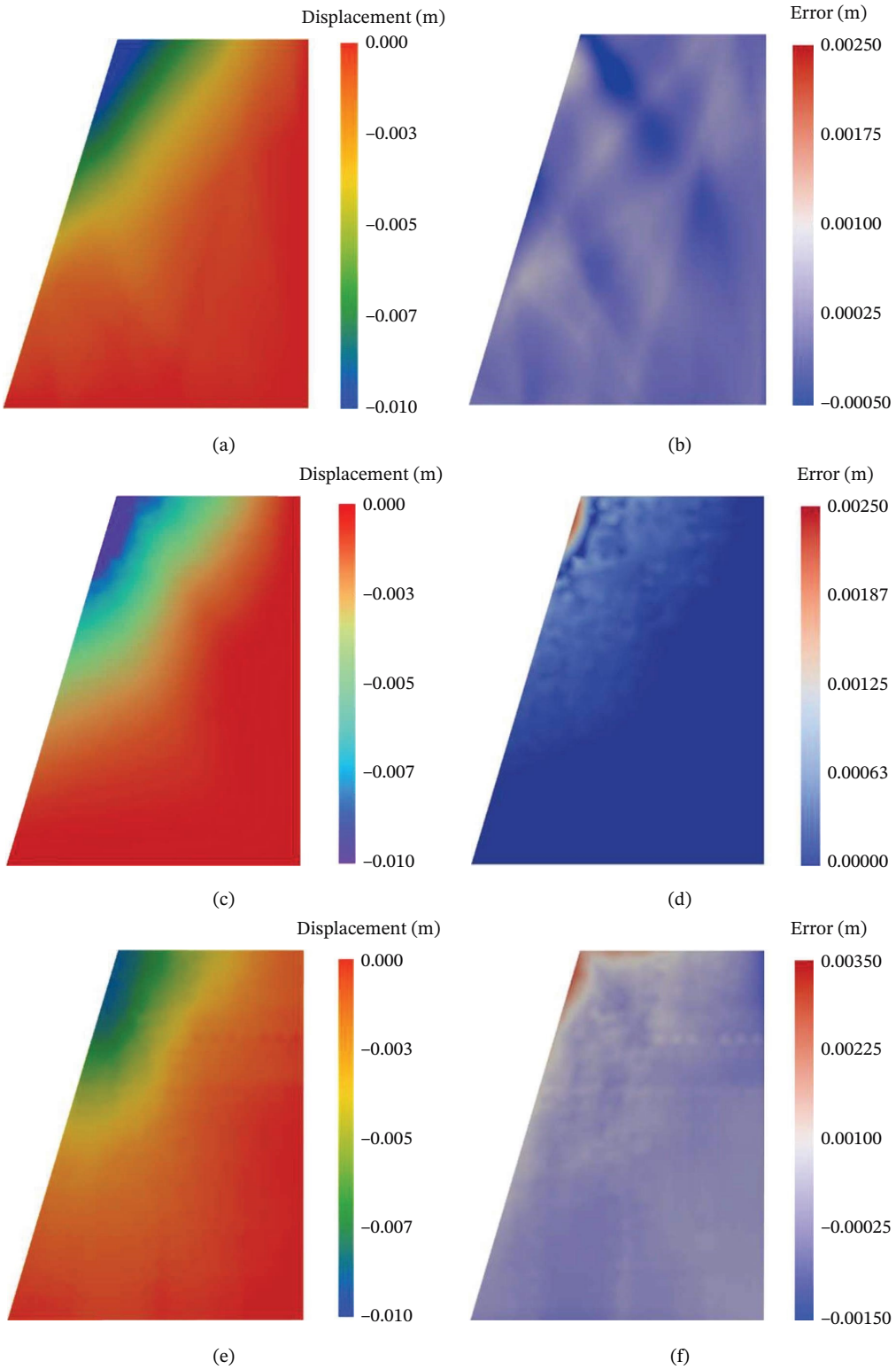


FIGURE 21 | Prediction results and error cloud map of the traditional models. (a) BP model prediction results. (b) Error cloud map for the BP model. (c) RF model prediction results. (d) Error cloud map for the RF model. (e) XGBoost model prediction results. (f) Error cloud map for the XGBoost model.

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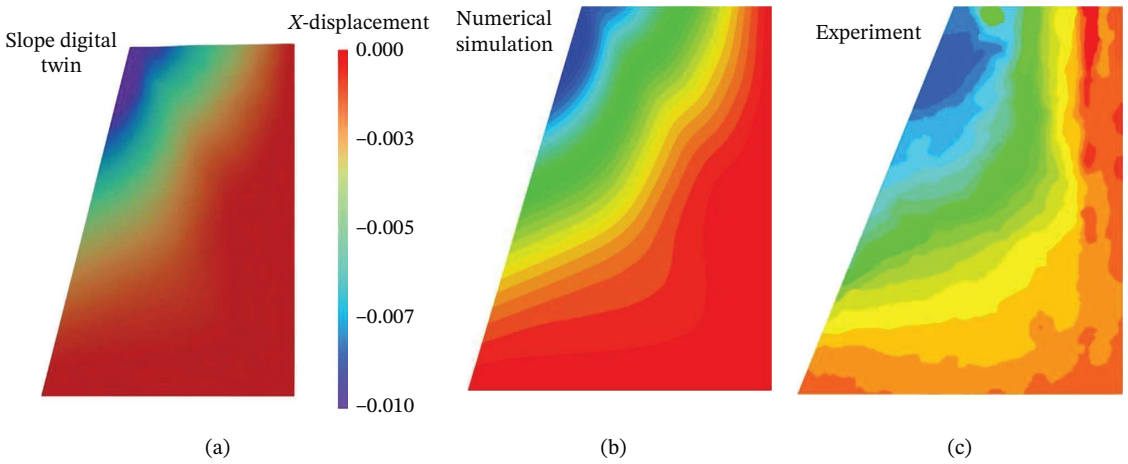


FIGURE 22 | Comparison of the contour maps of slope displacement in the *x*-direction. (a) Surrogate model results. (b) Numerical simulation results. (c) Experimental results.

4.4 | Surrogate Model Prediction Results

The monitoring point data are input into the pretrained surrogate model, and the predicted horizontal displacement values of the four monitoring points are extracted through the virtual monitoring points at the same location, as shown in Figure 17. The predicted displacement of all monitoring points tended to increase, but the predicted value fluctuated greatly. Compared with the experimental data, the predicted value is generally low: the error of monitoring point 1 is approximately 2 mm, and the relative error is approximately 25%. The error of monitoring point 2 is approximately 1 mm, and the relative error is approximately 20%. The error of monitoring point 3 is approximately 0.4 mm, and the relative error is approximately 20%. The error of monitoring point 4 is the smallest, approximately 0.1 mm, and the relative error is approximately 10%.

The optimised surrogate model is used to predict the horizontal displacement of the virtual monitoring point, and the results are shown in Figure 18. The prediction stability and accuracy of the optimised model are significantly improved. A comparison between the prediction results of the four monitoring points and the experimental data is shown in Figure 19. Specifically, the optimised prediction trend of monitoring point 1 is more consistent with the experimental data, and the error is significantly reduced. The error between the predicted value and the test value of monitoring point 2 remains stable at approximately 0.3 mm (relative error of 5%), which is within the acceptable range and significantly improved compared with that before optimisation. When the load of monitoring point 3 is small, the two predictions increase, but with increasing load, the prediction results of the optimisation model are closer to the actual data. Because the displacement value of monitoring point 4 is small and the change is minor, the optimised prediction is more accurate.

4.5 | Visualisation Results and Discussion of the Digital Twin Model

On the basis of the developed visualisation platform, the prediction results and numerical simulation results of the surrogate model based on integrated stacking learning, the prediction results of the traditional models (the RF, BP and XGBoost models are taken as examples) and the laboratory experimental results are compared and analysed systematically. The performance of the model in this paper is comprehensively evaluated from the perspective of the spatial distribution and cloud map of the displacement error in the *x*-direction.

The comparison between the cloud map of displacement in the *x*-direction (Figure 20(a)) generated by the surrogate stacking model and the numerical simulation results (Figure 20(b)) is shown in Figure 20. The two are highly consistent. The cloud map of the error is shown in Figure 20(c). The maximum error is 0.5 mm, the minimum is 0.18 mm, and the average error is 0.02 mm, indicating high prediction accuracy. The Relative Percentage Error (RPE) distribution is further illustrated in Figure 20(d). In the primary sliding zone, the relative error is maintained within 4%. Although the relative error appears higher in the stable region due to the small denominator effect (where actual displacement approaches zero), the absolute error remains negligible and does not affect the overall safety assessment.

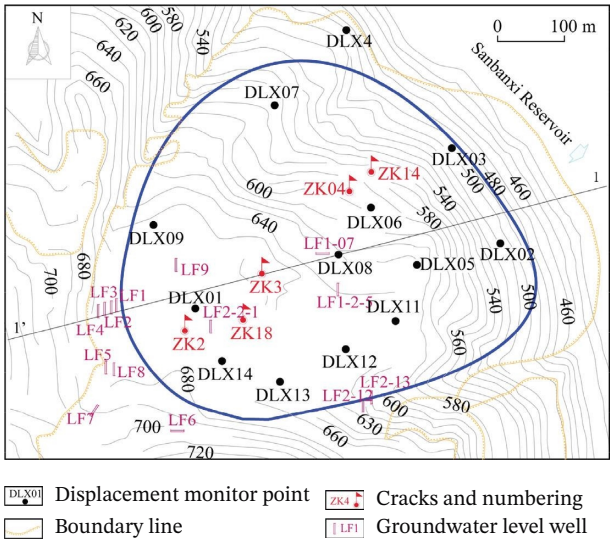


FIGURE 23 | Geological plan of the Donglingxin landslide project.

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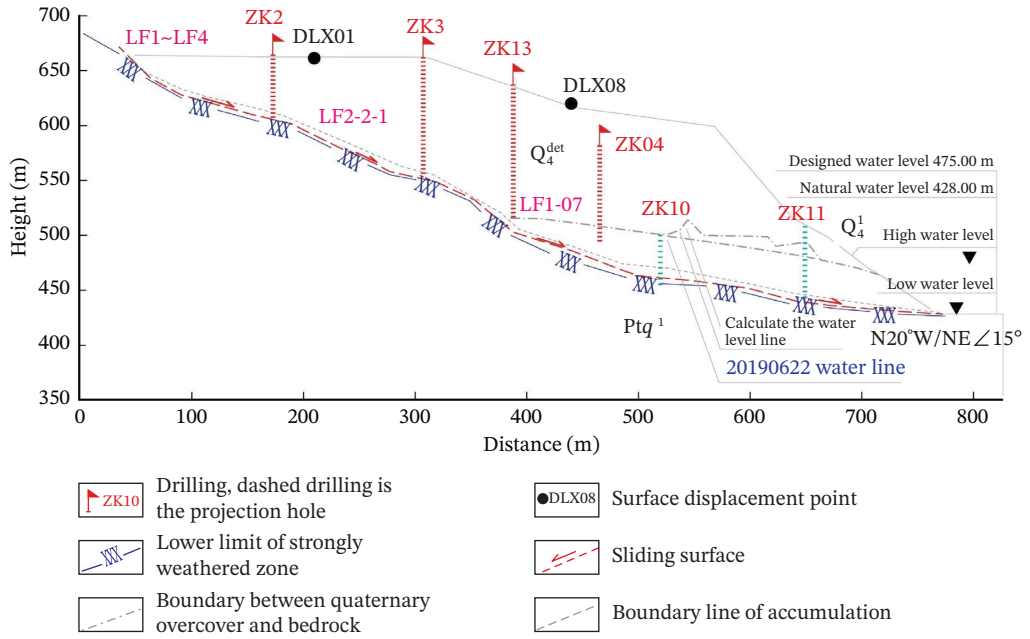


FIGURE 24 | Profile of the Donglingxin landslide.

To highlight the advantages of the integrated stacking method, the prediction nephograms and corresponding error nephograms of the BP model, RF model and XGBoost model under the same working conditions are compared in Figure 21. Among them, the BP model has the largest prediction error, with a maximum error of 1.6 mm and an average value of 0.2 mm; the RF model has second-largest prediction error, with an average error of 0.07 mm; and that of the XGBoost model is relatively small, with an average error of 0.05 mm, which is less than that of the stacking model.

Although the surrogate model was optimised, the overall accuracy of the digital twin model still largely depends on the quality of the training data. As shown in Figure 22, the error of the results of the digital twin model (Figure 22(a)) is less than that of the numerical simulation (Figure 22(b)), but the error between the digital twin model and the indoor experiment (Figure 22(c)) is greater, which is due to the simplification of the constitutive model, parameter uncertainty and the inherent difference between the simulation and the experiment in the numerical simulation.

5 | Engineering Application

5.1 | Engineering Background

In this paper, the proposed method is applied to the Donglingxin landslide [34], which is located in the reservoir area of the Sanbanxi hydropower station in Qiandongnan Prefecture, Guizhou Province. The landslide volume is approximately $2 \times 10^7\ m^3$, and the main sliding direction is N55°E. The planar area is approximately $2.91 \times 10^5\ m^2$; the landslide is characterised as an "armchair-like" landform with a width in the northeast and a width in the southwest. Its geological engineering plan is shown in Figure 23.

In accordance with the planar geological map and drilling data, a section of the landslide is drawn, as shown in Figure 24. The figure shows the main structural components of the landslide mass, including the sliding mass, sliding belt, sliding surface, sliding bed and other elements. The sliding mass is composed mainly of gravel and block stone, with obvious stratification characteristics from top to bottom: the upper part is a silty clay layer containing gravel and block stone, the middle part is transitional to a silty clay layer or cataclastic rock layer containing gravel, and the lower part is a silty clay layer containing gravel. The sliding zone is distributed at the bottom of the landslide accumulation body and above the sliding surface, and the lithology is mainly greyish brown, greyish yellow, yellow and grey gravelly silty clay. The sliding surface is composed mainly of muddy sandy gravel rock samples or gravelly clay, with a maximum burial depth of 145 m. The bedrock of the sliding bed is mainly thick massive metamorphic tuffaceous sandstone, banded tuffaceous silty slate and medium-thick layered metamorphic tuff.

On the basis of the topographic map of the Donglingxin landslide mass and in combination with the geological plan, section and

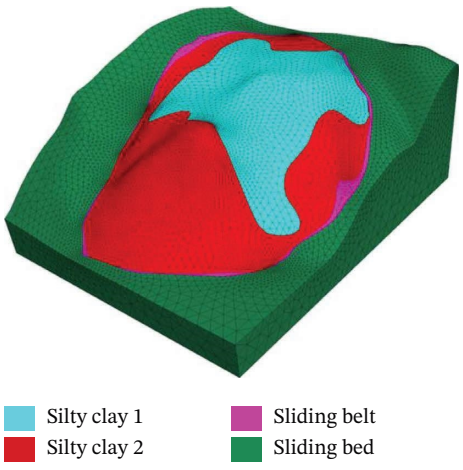


FIGURE 25 | Numerical simulation of the landslide.

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TABLE 7 | Rock and soil parameters.

Different zones	Density (kN/m^3)	Elastic modulus (MPa)	Poisson's ratio	Cohesion (kPa)	Friction angle (°)
Silty clay 1	20.5	100	0.35	17	22
Silty clay 2	22	500	0.25	140	25
Sliding belt	20	100	0.35	20	18
Sliding bed	25	10,000	0.2	9000	50

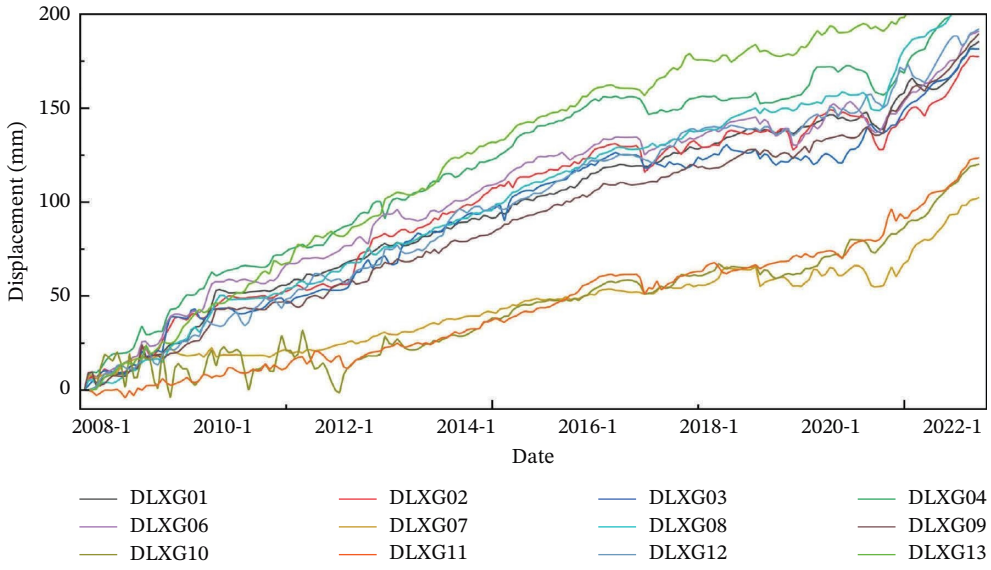


FIGURE 26 | Surface deformation monitoring data for the Donglingxin landslide.

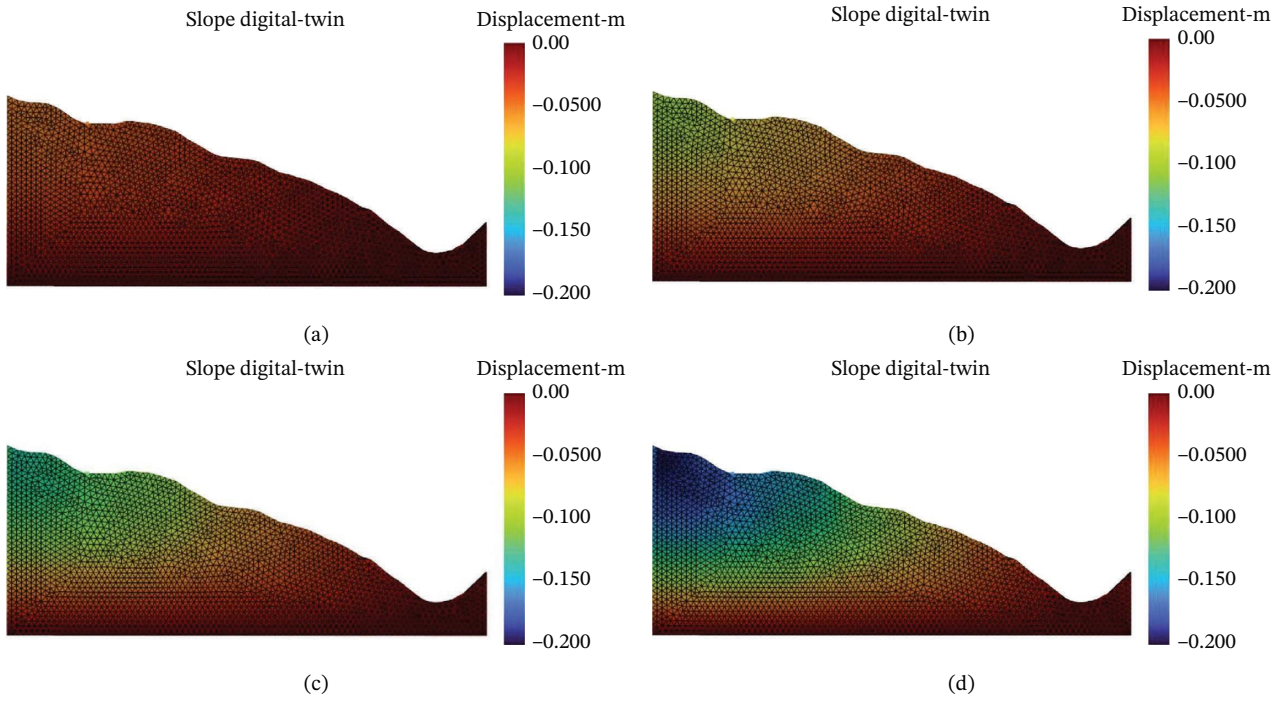


FIGURE 27 | Real-time displacement cloud map for the Donglingxin slope. (a) Displacement cloud map for 2010. (b) Displacement cloud map for 2014. (c) Displacement cloud map for 2016. (d) Displacement cloud map for 2021.

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borehole survey data, a three-dimensional numerical analysis model is established, as shown in Figure 25, and the rock and soil parameters are determined, as shown in Table 7.

5.2 | Landslide Monitoring

The natural water level of the Donglingxin reach of the Sanbanxi hydropower station is 425 m, and the water level in the design is 475 m. The reservoir has been impounded since January 2006. From April to July 2007, the reservoir water level in this reach rose rapidly from 430 to 470 m, an increase of 40 m in three months, with an average daily rate of increase of approximately 0.5 m. In late July 2007, cracks appeared on the surface at the rear edge of the reservoir bank slope, accompanied by signs of geological disasters such as wall dislocation and house tilt. Since 2008, a systematic geological survey has been carried out on the slope of the reservoir bank, and deformation monitoring began in 2009. The layout of the monitoring points is shown in Figure 23. A total of 13 monitoring points are established, of which DLXG04 is located outside the landslide, and the remaining 12 points are located on the surface of the landslide. The monitoring results are shown in Figure 26.

5.3 | Landslide Monitoring Results

On the basis of the digital twin model and real-time monitoring data of the Donglingxin landslide, a displacement nephogram of the landslide is generated, as shown in Figure 27. The slope displacement nephogram in 2010 is shown in Figure 27(a), with a maximum displacement of approximately 54 mm, reflecting the initial deformation characteristics of the landslide; Figure 27(b) shows the cloud map for 2014, with a maximum displacement of approximately 103 mm and a significant increase in displacement within four years, with a growth rate of approximately 12.25 mm/y; Figure 27(c) shows the 2016 cloud map. The maximum displacement increased to 134 mm, indicating that the deformation rate further increased, reaching 15.5 mm/y, and the landslide entered the accelerated development stage; Figure 27(d) shows the cloud map for 2021, with the maximum displacement reaching 193 mm, revealing that the landslide displacement continued to increase, but the growth rate slightly decreased to 11.8 mm/y. This series of cloud maps clearly shows the deformation law of the Donglingxin landslide over more than 10 years: the displacement increases annually, and the rate of increase first increases but then decreases with time. The results show that the Donglingxin landslide experienced "acceleration–stabilisation" and entered a relatively stable deformation stage.

6 | Conclusion

In this paper, a digital twin modelling method based on an optimised surrogate model is proposed for slope monitoring; it achieves accurate real-time mapping and visualisation of the slope state driven by data from limited monitoring points and effectively considers the calculation efficiency and response accuracy of the digital twin system. The main conclusions are as follows:

1. To address the problems of high computational complexity and poor real-time performance of digital twin models of slopes, a surrogate model based on stacking-based ensemble learning is constructed. On the training set, the goodness of fit R^2 of the model is 0.971, and the RMSE is 1.511; the R^2 on the verification set, it is 0.955, and the RMSE is 2.007; and on the test set, the R^2 is 0.952, and the RMSE is 2.108, indicating excellent prediction performance and generalisability. Furthermore, a dynamic optimisation mechanism that integrates multidimensional error analysis, adaptive parameter adjustment and error sample weighting is proposed to achieve continuous performance improvement of the surrogate model in practical applications.
2. Through an indoor slope loading failure experiment combined with DIC technology, high-precision displacement field and cloud image data of the entire process of slope deformation are obtained. After optimisation, the prediction error of the surrogate model at the monitoring points is significantly reduced from the original 20% to 5%, which verifies the effectiveness of the proposed optimisation strategy in improving the prediction accuracy of the model.
3. The prediction results of the surrogate model based on stacking-based ensemble learning are compared with the numerical simulation results, traditional model prediction results and laboratory experiment results. The maximum error of the prediction results of the stacking model is 0.5 mm, and the average error is 0.02 mm, which is better than those of the XGBoost, RF and BP models. However, the prediction results still differ from the actual experimental results because of the simplification of the model by numerical simulation and the uncertainty of the parameters.
4. A digital twin system for a slope that integrates physical monitoring and virtual prediction is developed, and the method is successfully applied to the actual Donglingxin landslide project, realising real-time dynamic visual monitoring and retrospective analysis of landslide displacement. The monitoring results reveal that the deformation rate of the landslide gradually increases from 12.5 to 15.5 mm/y and then decreases to 11.8 mm/y, revealing a dynamic "acceleration−stabilisation" process and that the landslide deformation has gradually entered a relatively stable stage.

The surrogate model-driven digital twin method developed in this study provides an efficient and reliable digital solution for slope engineering safety monitoring and early warning of disasters, and it has good theoretical significance and engineering promotion value.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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